

# HW18 - Haptic Paddle Submission

This report includes the required normalized force-displacement design, mechanical drawing, block diagram, and circuit diagram for the haptic paddle. The code files in the same folder read the AS5600 paddle angle, update computer graphics, calculate desired current, and send the current command over USB serial.

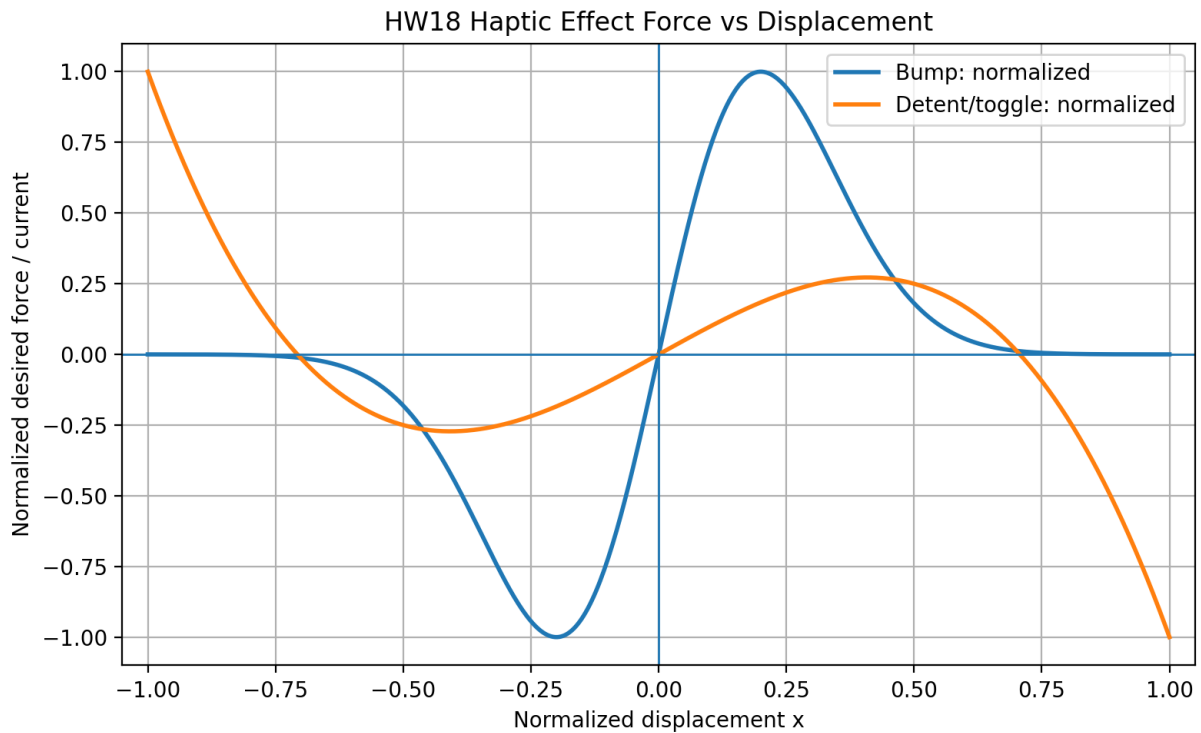
## Haptic Effect Equations

Normalized displacement:  $x = \text{angle\_deg} / 180 - 1$ , clipped to  $[-1, 1]$ .

Bump effect:  $F_{\text{bump}}(x) = A(x - x_0) \exp(-((x - x_0)^2)/(2 \sigma^2))$ , normalized to  $\pm 1$ .

Detent effect:  $F_{\text{detent}}(x) = kx - bx^3$ , normalized to  $\pm 1$ .

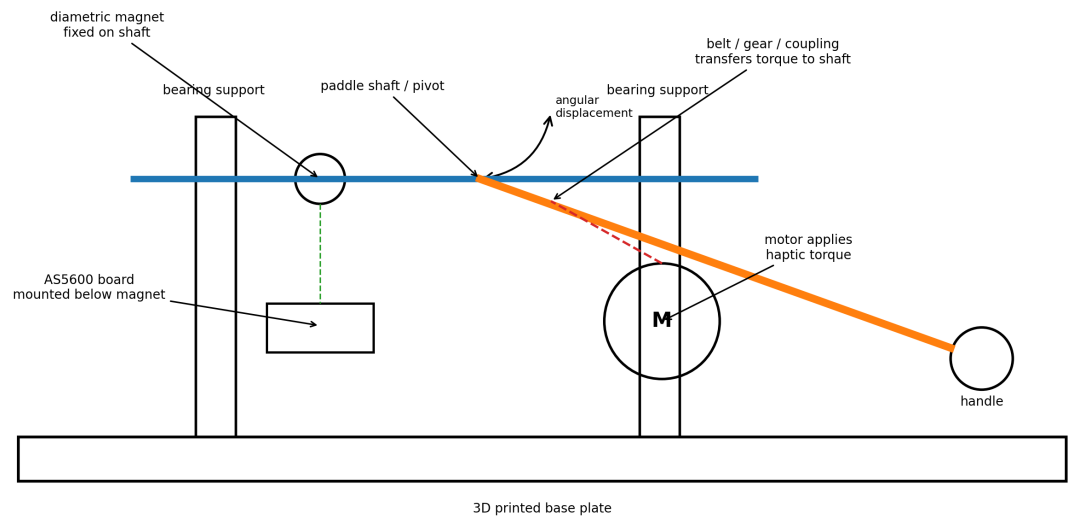
Desired current:  $I_{\text{des}} = I_{\text{max}} * F_{\text{normalized}}$ , with  $I_{\text{max}} = 0.30$  A by default.



# Mechanical Parts Sketch

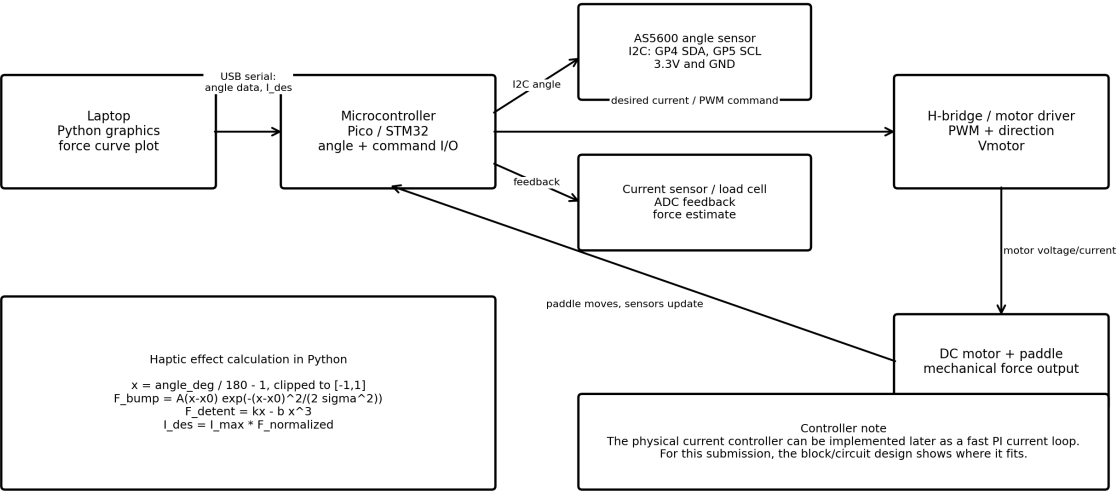
## HW18 Mechanical Parts Sketch

The paddle rotates about a shaft. A magnet on the shaft is read by the AS5600. The motor applies torque to create bump/detent effects.



# Block Diagram

HW18 Haptic Paddle Block Diagram



# Circuit Diagram

HW18 Haptic Paddle Circuit Diagram

Pins match the code for AS5600 I2C: GP4=SDA, GP5=SCL. Motor driver pins are a proposed controller stage for desired current/PWM output.

